

00 - Spec Base: Environment Constraints & Locked Decisions

Package: Bhadravati Budget Build Spec (agent-researched buy list)

Date: 2026-08-22 - **Status:** DRAFT v0.1 - research tracks merging

Relationship: extends Bhadravati_Interior_Design_V2/ governance. Changes NO locked decision. Fills the "street prices + exact SKUs" gap left open in FINAL_DELIVERABLE/06_Budget_Phasing.md ("Rs bands to be filled after local quotes").

1. Site environment (design drivers)

Constraint	Source	Design response
Fine dust + soot deposition (industrial belt)	Client brief + SoT problems-solved	Mid-tone washable walls (no stark white); closed storage; sealed windows; low-pile textiles; purifier
Smoke smell / oily deposits	Client brief	Washable emulsion class >=5000 scrubs; degrease-clean cadence; exhaust strategy review
Noise day/night	Client brief	Window seals + curtains + soft furnishings; realistic dB expectations documented
Vibrations (factories/trucks)	Client brief	Anti-vibration pads under appliances; rigid wardrobe fixing; no floating shelves over bed
Monsoon humidity ~85%	Climate knowledge	BWP/710 boards at wet risk, E1/E0 emission boards, matte finishes (hides water haze), ventilation
Hard water	SoT maintenance notes	Wipe-dry granite discipline; geyser tank chemistry choice; matte shutters not gloss

2. Locked decisions carried forward (from V2 locks)

Slot	Lock	Value
Walls	Birla Opus NN9074 Puddle of Grey	#B5AB9C warm greige, low-sheen washable
Ceiling	WW0005 White Linen	#EEEEDE9
Accent (optional, <=1 wall)	GG7140 Tender Buds	#9CAE91 sage
Kitchen shutters	Century StarLine S1241 MT Latte	existing black granite retained; shutters only
Wardrobe shutters	Century 84689 SU Idria Oak	3 doors, leaf 457/457/458 mm in 1372 mm clear x 2286 H x ~488 D
TV cabinet accent	Century 80236 DW Slate Grey	controlled low accent only
Hardware	brushed stainless recessed pulls	SS304 screws at wet risk
Lighting	3000 K baseline, CRI>=90 where colour matters	matte diffusers
Banned	gloss/sparkle, gold strips, TV feature wall, island, L-flip, slat props	-

All codes verified against processed_pdf_text/birla_opus_shade_card_extracted.txt (2,322-shade card, compiled 03-Aug-2026) and processed_pdf_text/century_starline_extracted.txt.

3. Brightness ladder for darker rooms (card-evidenced)

The client's own shade card recommends for **industrial areas**: ceilings WW0005/WW0020, walls NN9088/WW0120/NN9242. Measured brightness (L*~ from RGB):

Code	Name	Hex	L* approx	Use rule
NN9074	Puddle of Grey	#B5AB9C	~71	LOCKED default - best soot-hiding mid-tone
NN9242	A Khadi Kurta	#C7BAB0	~77	Factory-recommended industrial wall shade - use instead of NN9074 in dim rooms (bath corridor, wardrobe surround)
NN9088	Ecru Tint	#E9E3D9	~89	Secondary/light option - stricter wipe schedule (shows dust faster)

Code	Name	Hex	L* approx	Use rule
WW0020	Virgin White	#EDE9E2	~92	Ceiling alternative to WW0005

Rule: physical samples approved under morning sun, afternoon sun, 3000 K evening before ordering (per 03_Palette_Materials.md metamerism warning).

4. Paint system product ladder (from card system notes - pricing/chemistry under research)

Tier	Product	Claimed class
Luxury	Birla Opus One Pure Elegance Matt	scuff & stain resistant, 7-yr warranty class
Premium	Birla Opus Calista Ever Wash	5000+ washes, anti-fungal
Economy	Birla Opus Style Color Smart	anti-fungal, washable
Prep	One Pro Putty+Primer / Calista Pro White Primer	-

Cross-brand comparison (Asian Paints Royale/Apex, Berger, JSW, Dulux) incoming from research track R-Paint.

Research tracks in flight

Track	Scope	Report file	State
R-Paint	Emulsions by series, VOC, scrub cycles, shades, human validation	07_paint.md	[OK] COMPLETE
R-Wood	Plywood/HDHMR grades, IS standards, formaldehyde classes, hardware, carcass spec	02_wood_joinery.md	[OK] COMPLETE
R-Light	LED fixtures/datasheets, wiring, MCB/RCCB/SPD surge plan, lux schedule	03_electrical.md	[OK] COMPLETE
R-Switch	Switchgear series/materials, BLDC fans, exhaust, geyser	04_utilities.md	[OK] COMPLETE
R-Envelope	Dust/noise/vibration defense, flooring, air purification	05_envelope.md	[OK] COMPLETE
R-Furniture	Sofa/mattress/bed/study/bedding by construction + real-user validation	01_furniture.md	[OK] COMPLETE

Consolidated priced buy list: 06_MASTER_BUY_LIST.md - Essential ~Rs 2.9-3.6L - Recommended ~Rs 3.9-4.9L - Premium Rs 5.5L+.

Shared citation ledger: 40 sources (~/.hermes/cache/citations/ledger.json), Sources block rendered into 06.

01 - Furniture, Mattress & Soft Goods (research track R-Furniture)

Status: COMPLETE - agent research recovered + direct re-verification. Prices EST 2026.

1. Sofa - by construction, not looks

Rung	Pick	Specs	Price
Budget Rs 12-15k	Local carpenter: kiln-dried/teak frame + custom cushions	spec ≥ 32 kg/m ³ seat foam (cheap 24-28 sinks in ~2 yrs), tight-weave performance fabric	Rs 12-15k
Recommended	Wakefit Napper Fabric 3-seater [3]	termite-resistant neem solid wood frame, polyester upholstery, 194x85x85 cm, foam fill, knock-down delivery+install, 3-yr warranty (frame/mechanism 3 yr; fabric/foam/sagging 1 yr) [6]	Rs 20,299 (Omega Grey Mini; velvet variants Rs 32.3k MRP-discounted)
Premium Rs 35k	IKEA MANNARP Gunnared beige [7]	washable cover (dust win), 10-yr warranty, high backrest	Rs 34,990

Real-user validation:

- Flipkart buyers (490 ratings, 4.4★): "really worth the money... fabric is very good in quality... no issues after a month" [4]
- (!) Counter-signal: one LinkedIn consumer post - Napper Plus fabric **faded in patches within a year indoors**, support initially dismissive [5]. Mitigation: pick mid-tone grey/beige (our palette anyway), avoid direct sun placement, keep warranty card + unboxing video.
- Dust logic applied: NO white bouclé, NO deep-tufted velvet (dust traps); tight-weave mid-tone wins.

2. Mattress - heat + durability first

Rung	Pick	Why	Price
Budget <Rs 10k	Kurlon Dual 4-5in flip	60-yr brand, Sheela Foam-owned, try in-store; 4in only if sleeper <75 kg [8]	Rs 7-10k
Value	Wakefit DreamPod / Sleepyhead BodyIQ (Duroflex-made, best South-India service)	HR foam breathes better than memory foam in warm rooms; 10-yr warranty class [8]	Rs 12-16k
Winner	SleepyCat Latex Hybrid 8in queen [8][9]	latex comfort layer = coolest material class, dust-mite resistant (industrial allergy bonus), HR base support, "best buy-it-for-life under Rs 20k", 10-yr warranty, 100-night trial	Rs 18-20k

Warranty fine print that matters (all brands): "10-year warranty" = full replacement year 1 only, pro-rated credit after; body impression must exceed 1.5in to qualify; one stain can void it -> use a waterproof protector from day 1 [8].

"Orthopedic" is an uncertified marketing word in India - medium-firm feel (6.5-7.5/10) is the actual medical guidance [8].

Latex longevity note: ask for GOLS certificate on latex layers, else it may be blended [8].

3. Bed frame

Solimo Petra solid sheesham queen (teak finish) - Rs 14.5-17.5k current, historically hits **Rs 11,999-14.5k** in sales (set price alert) [10]. Solid wood > engineered for creak resistance under vibration; metal frames squeak without rubber-washer bolts and rust near bathrooms [11]. **Skip box storage** - under-bed cavities collect industrial dust and are never cleaned; choose clean-leg design, vacuum underneath weekly with robot/manual.

4. Study setup $\leq$ Rs 15k

- **Chair: Cellbell Desire C104 mesh** - Rs 2.9k EST; 105 kg rating, metal base, breathable mesh (sweat-free in warm climate), tilt-lock; validated across buyer guides as best strict-budget pick [1][2][13]. Upgrade path later: Da URBAN Merlion / Green Soul Jupiter-class when budget allows.
- **Desk: engineered-wood top + metal legs** (1200x600 mm) or carpenter-built matte laminate top matching palette - Rs 3.5-6k.
- Total ~ Rs 7-9k, leaving headroom for the task lamp in 03_electrical.md.

5. Coffee table & TV unit (size/material confirmation only)

- Coffee table: round matte Ø700-800 x H400-450 mm - engineered top or solid rubberwood, matte finish per locked spec; ready-made Rs 2-4k or in wardrobe-carpenter scope.
- TV unit: floating 1400-1800 mm slate-grey laminate - **carpenter-build alongside wardrobe/kitchen** (same boards, edges, labour mobilization = cheapest route); ready-made equivalents don't match the locked 80236 DW finish reliably.

6. Bedding

100% cotton fitted sets, 200-300 TC, warm neutrals (taupe/greige): Solimo fitted sets hit **Rs 626-920** on sale cycles [12]; buy 2-3 sets (weekly wash cadence in this environment). Avoid microfiber (static attracts dust).

[1] greensoul.online brand comparison - [2] ergochairs.in Seoul-X analysis - [3] Flipkart Napper listing - [4] Flipkart Napper reviews - [5] LinkedIn consumer post Naresh V - [6] pricehistory.app Napper spec/warranty - [7] ikea.com MANNARP - [8] readthisfirst.in mattress guide 2026 - [9] dealzza.in mattress guide - [10] cheapestinindia.com Solimo Petra history - [11] republicworld.com wood vs metal beds - [12] cheapestinindia.com Solimo bedsheets history - [13] naatural.in WFH chair guide

02 - Wood, Joinery & Wardrobe Hardware (research track R-Wood)

Status: COMPLETE - prices are 2025-26 street EST - confirm at Shimoga/Bhadravati dealer.

Locked context: wardrobe niche 1372 W x 2286 H x ~488 D mm; 3 doors (457/457/458 leaves); shutters in Century 84689 SU Idria Oak; kitchen = shutters-only over existing granite; recessed SS pulls.

1. Board decision matrix

Grade / standard	Use here	Brand pick	Rs /sqft 19mm EST	Warranty	Notes
BWP IS:710 (boiling-water-proof)	Kitchen carcass+shutters core (wet risk), bathroom-adjacent wardrobe wall	Century Sainik 710 [14][15]	Rs 140-165 [16]	8-10 yr termite/borer + BWP bond	"most recommended BWP in Bangalore contractor community" [15]; Reddit-validated go-to
BWP premium	only if budget allows	Greenply Gold / Century Club Prime	Rs 185-205 [16]	30 yr; Gold = CARB + zero-E formaldehyde	worth it for bedroom health if splurging
MR IS:303	dry interior shelving	Sainik MR / Ecotec	Rs 84-109 [16][14]	5 yr	NOT for monsoon-humid wet walls
HDHMR >=850 kg/m3	shutter cores alternative, loft band	Action TESA	Rs 95-150 [17]	brand-varies	flat, screw-holding, industry default for laminate shutters
Particle board	■ avoid	-	Rs 55-80	-	swells on water contact; short-life builds only [17]

Cheapest-sensible combo (spec):

- Kitchen: carcass + shutters **Sainik 710 19mm**, loft band may be HDHMR if sealed edges.
- Wardrobe: carcass **Sainik 710 or MR ply on the dry side**, shutters **HDHMR 18mm** (flatness for Idria Oak grain) - or full 710 if quotes converge.
- Internal shelves: 12mm ply; back panel 6mm ply rabbet-fixed (not MDF).
- Chemistry notes: BWP = phenolic resin bond survives boiling water 72h (IS:710 test); termite treatment = boron-based salts in core glue-line; prefer E1/E0 emission class or Greenply Gold CARB for bedroom air quality.

2. Finish layer beyond locked Idria Oak

- Solids to pair: Century StarLine solids or Merino/Greenlam matte solids in greige/taupe family (~Rs 900-1400/sheet EST). Match undertone to Latte (#A49483-ish family), matte texture only.
- Edge banding: **PVC 1mm wardrobe / 2mm kitchen**, hot-air applied, ABS preferred (E3 EBT already sourced locally per SoT). Laser edge = premium overkill here.

3. Hardware picks (the pro's split strategy)

Pro strategy from trade sources: **spend where hands touch daily - Hettich in kitchen, value-brand elsewhere** [19]:

Item	Pick	Price EST	Why
Kitchen hinges (soft-close)	Hettich Sensys 8645i (110°, integrated damper, 5-40°C rated) [18]	Rs 257/set of 2 (Rs 515 listed pair) [18]	EN 15570 L3 durability; Indian-climate rated
Wardrobe hinges (soft-close)	Ebco Euro Slow Motion	Rs 120-180/pc EST	ISO/BIFMA/BIS certified, reliable [19]
Drawer channels	Hettich Quadro (kitchen) / Ebco telescopic (wardrobe)	Rs 900-1600/pr vs Rs 350-600/pr	budget split same logic
Pulls	recessed brushed-SS J-pull (LOCKED)	Rs 80-150/pc	no bar handles
Screws wet zone	SS304	-	anti-rust mandate

Anti-fake note: buy hinges/channels from authorized dealers (HomeRun-type aggregators publish dealer-sourced pricing [29]); check Hettich laser etch + packaging QR.

4. Carcass construction spec (3-door wardrobe)

- Leaves: 457/457/458 x 2286 mm - vertical Idria Oak grain, 18mm HDHMR/ply core, PVC 1mm edges both faces.
- Hanging rod height: 1000-1100 mm AFF for shirts; top shelf >=1800 mm for infrequent; drawers bottom zone <=600 mm reach.
- Back: 6mm ply in rabbet + screwed to wall studs @ 400mm centers - **anti-vibration**: use 2-part chemical anchor or nylon sleeves so metal-to-wall contact doesn't transmit factory vibration as buzz.
- Mirror: 4mm with safety film (IS 2553 class) inside a leaf.
- LED sensor strip: optional later - wiring conduit during build even if light deferred.

5. Human validation

- Contractor community consensus: Sainik 710 = value king for BWP; multiple Reddit renovation threads cite it post-comparison [15].
- Trade cost guide: branded soft-close hardware upgrade (Rs 5-15k over local) is "the single best-value line item" and adds 4-6 yrs life vs cheap hinges that sag in 3-5 yrs [20][21].
- Full wardrobe market rates for cross-check when carpenter quotes arrive: openable laminate mid-spec Rs 1800-2600/sqft of front area installed (2026) [17]; our 70 sqft front ~ Rs 1.26-1.8L all-in sanity band.

[14] setkar.in CenturyPly price list 2025 - [15] home-run.co plywood comparison - [16] hdhmr.in price list - [17] studiomatrix.org wardrobe cost guide - [18] hardwaredepot.in Sensys 8645i - [19] home-run.co hinge comparison - [20] interiorcalculator.com hinged wardrobe - [21] designcafe.com material costs

03 - Lighting & Electrical Infrastructure (research track R-Light)

Status: COMPLETE - 3000 K locked; industrial-grid surge priority.

Datasheet-grade picks below; prices street EST 2026.

1. Fixture picks

Slot	Hero pick	Specs (datasheet)	Budget alt
Downlights (living/bed/kitchen amb.)	Philips Meson Comfort 13W/17W recessed, 3000 K [24]	CRI$\geq$90, 30,000 h, interlaced optics (~40% glare), EyeComfort, 3-yr warranty; 13W=1030 lm ($\varnothing$ 125), 17W=1360 lm ($\varnothing$ 150)	Havells Luna JB 12W - Rs 215, 90 lm/W, 3 kV surge + 380 V protection , PC housing [23]
Bath fixture	Havells Luna-class IP44 or bulkhead with same surge spec	IP44 min, 3000 K	-
Kitchen task	3000 K under-loft strip, diffused aluminium profile, CRI $\geq$ 90 (Meson-class module or Wipro Immedio-class)	106in run ~ 2700 mm; ~1200-1500 lm total	generic 24V strip + branded driver
Study lamp	3000 K/4000 K switchable LED task lamp	500 lux at desk	-
Bedside	2x 2700-3000 K table lamps	separate switching	-

Why Meson Comfort as hero: CRI 90 renders Latte/Idria/granite true (locked requirement), 30k-hr life, glare control in a low-ceiling studio. **Why Luna as budget alt:** 3 kV surge protection is rare at Rs 215 and this is an industrial grid - every fixture should carry surge tolerance $\geq$ 2.5 kV.

2. Cove lighting verdict (honest)

Skip coves. In constant-dust air, horizontal cove shelves collect soot film within weeks and need ladder-cleaning monthly; the "premium feel" gain is small vs maintenance pain. Spend instead on CRI-90 downlights + one quality floor/table lamp for evening layering. (If insisted: sealed-profile 24V strip with end-caps, never open LED tape.)

3. Wiring & protection plan (studio-sized)

Item	Pick	Spec	Price EST
House wire	Polycab FRLS 1.5 sqmm (lights) / 2.5 sqmm (power)	99.97% Cu, FRLS low-smoke	Rs 1,638/roll(90m) Bengaluru trade [27]; 2.5 sqmm ~Rs 2.6-3k/roll
RCCB	Schneider Easy9 EZ9R35225 25A 2P 30mA AC-type	whole-studio shock/fire protection	Rs 1,786 [25]
SPD	Schneider Easy9 EZ9L33620 Type-2, 1P+N, In 10kA / I_{max} 20kA, Up 1.3kV [26]	"medium risk" class - right for industrial belt; pair with 25A C-curve MCB	~Rs 2-3.5k EST (dealer)
DB	8-way double-door MCB box (Schneider/Legrand)	house RCCB+SPD+6 SP MCBs	Rs 1.2-2k
MCBs	6A lights x2, 16A power x2, 16A geyser, 20A AC-ready, 25A C-curve (SPD backup)	-	Rs 150-250 ea
Earthing	verify/refresh pit + bonding to panel	licensed electrician mandatory	site

Circuit map: L1 lights (living+study), L2 lights (bed+bath+kitchen), P1 sockets living/study, P2 sockets bedroom/kitchen, G geyser 16A, AC-ready 20A, SPD on incomer.

4. Room lighting schedule (3000 K throughout)

Zone	Target lux	Fixtures
Living	150-200 ambient	4x Meson Comfort 13W (~4100 lm total)
Kitchen counter	400-500 task	under-loft strip 1500 lm + 2x 9W downlights aisle
Bedroom	50-150 ambient + bedside task	2x 9W downlights + 2 table lamps
Bath	150-300, IP44	1x 12W + mirror-flank light

Zone	Target lux	Fixtures
Study desk	500 task	task lamp + 1x 9W general
Wardrobe interior	colour-true	3000 K sensor strip (conduit now, light optional)

5. Human validation

- 14-yr electrical shop log: in voltage-spike areas, **Wipro Garnet showed slightly fewer warranty claims over 24 months** than Havells on bulbs; both handle 270V+ surges; Havells wins warm-white batch consistency (2700K) - relevant to our 3000K/2700K lamp picks [22].
- Luna-class 3 kV protection + 380 V high-voltage cutoff is the spec to demand in any budget fixture for this grid [23].

[22] aarushielectrical.com Havells vs Wipro 2026 - [23] havells.com Luna 12W page - [24] Philips Meson Comfort catalogue PDF (npcdn) - [25] Flipkart Easy9 RCCB EZ9R35225 - [26] se.com EZ9L33620 - [27] IndiaMART Polycab FRLS Bengaluru

04 - Switches, Fans & Utilities (research track R-Switch)

Status: COMPLETE - prices street EST 2026.

1. Modular switches

Brand / series	Positioning	Price signal	Verdict
Legrand Myrius NextGen	premium-value leader in 2026 buyer guides [29]	Rs 55/pc trade (6A, Bengaluru wholesale) [28]; retail Rs 111+	WINNER - best finish-per-rupee, silver-alloy contacts, PC housing
Anchor Roma Urban / Tresa	value mainstream	~40-70% of Myrius NextGen per HomeRun comparison [29]	runner-up ultra-budget
Schneider Livia/Opus	mid-premium	comparable to Myrius	fine but no edge here
GM/Havells Reo/Fabio	mid	-	acceptable; weaker resale of spare modules
Vinay/Kundan	ultra-budget	-	only for hidden utility points

Materials notes: look for **silver-alloy contacts** (arcing resistance), brass terminals (phosphor-bronze springs on better series), **UL94 V-0 polycarbonate** housing (self-extinguishing) - all true of Myrius NextGen class.

Anti-fake: buy from authorized dealers (HomeRun/brand stores publish dealer-sourced live pricing [29]); check moulded logo depth + GST invoice.

2. Ceiling fan

Atomberg Efficio Alpha 1200 mm BLDC - Rs 2,699 [30][31]

- 28 W (vs 75 W induction), BEE 5-star, 220 CMM air delivery, IR remote + LED speed indicator.
- **Gloss finish specifically reviewed as the right choice for dusty cities** ("matte fans show fingerprints and dust streaks within weeks... Efficio stays looking clean with a quarterly wipe") [30].
- Real users: strong airflow at low watts, quiet, runs well under low-voltage conditions (industrial grid relevant) [31].
- Register within 30 days for the extended 1+1 yr warranty [30].

3. Ventilation strategy for dust + smoke

Honest engineering take: **exhaust-only is wrong here**. Running exhaust fans in dust/soot-laden air depressurizes the flat and pulls more dusty air IN through every gap. Strategy:

1. Bath exhaust: yes (moisture must exit) - 150 mm with **gravity shutter flap** (stops backdraft dust), Havells/Crompton class Rs 800-1400 EST.
2. Kitchen: rely on chimney/recirculating mode or window-positioned exhaust used *only while cooking*, flap closed otherwise.
3. Fresh air: windows sealed + filtered (see 05_envelope.md), not unfiltered infiltration.
4. Chimney: budget auto-clean 60cm (Faber/Elica/Glen entry models Rs 8-12k EST) if cooking is oily - it protects Latte shutters and NN9074 walls from soot film; highest-leverage kitchen appliance after the hob itself.

4. Water heating (hard water winner)

Rung	Pick	Price EST	Why
Winner	AO Smith SDS/SGS-GREEN-015 15L - Blue Diamond glass-lined tank, glass-coated Incoloy element, Mg anode, 7-yr tank warranty	Rs 6,599 (2kW VCS-Pro class) to Rs 11,499 (3kW SGS) [32][33]	"no other geyser matches" in hard-water durability tests; scale-free element after sustained testing [32]
Budget	Crompton Arno Neo 15L (Nano Polybond)	Rs 5,999 [33]	best cheap performer; shorter element life in very hard water
Note	2 kW works on existing wiring; 3 kW needs dedicated 16A line	-	our DB plan already reserves geyser circuit

Maintenance cadence for hard water: annual descale flush, anode check every 2-3 yrs -> 8-12 yr life even in hard-water zones [33].

5. Small utilities

- Doorbell: wireless battery chime (Rs 400-700).

- Night lamps: 0.5W LED (Rs 100 ea) in bath corridor + study path.
- USB modules: skip wall USB-A modules (obsolete standard); one 20W GaN charger per desk/bedside instead.
- Extensions: 2x branded spike guards with surge clamping for TV/desk clusters.

[28] IndiaMART Legrand Myrius NextGen Bengaluru - [29] HomeRun modular switches price list 2026 - [30] smarthousegears.com Efficio Alpha review - [31] atomberg.com product page + reviews - [32] geyserslab.in 15L 5-star comparison - [33] winnerwinnerchickendinner.in 15L geysers under Rs 15k

05 - Envelope Defense: Dust, Smoke, Sound, Vibration + Flooring (track R-Envelope)

Status: COMPLETE - strategy: seal tight, filter what enters, absorb what remains.

1. Window sealing ladder (cheapest -> best)

Rung	What	Cost EST	Effect
1	EPDM rubber + brush pile seals on aluminium slider meeting faces & frame gaps	Rs 30-60/ft material	kills whistling gaps, big dust reduction
2	Neutral-cure MS-polymer sealant on perimeter gaps (paintable, stays flexible, low VOC vs silicone which off-gasses acetic acid & never paints)	Rs 350-500/tube, 2-3 tubes	permanent air-seal
3	Latch/compression upgrade + door bottom sweeps	Rs 200-400/door	presses seals shut
4	Secondary glazing: acrylic sheet insert on clip/L-frame inside reveal (poor-man's double glazing)	Rs 180-300/sqft DIY	real dB cut on traffic noise + dust barrier; validated by DIY builders (r/DIY acrylic insert projects report major street-noise drop) [38]
5	Full DGU aluminium window replacement	Rs 650-1100/sqft installed	last resort - 10x rung-1 cost

Verdict: do rungs 1-3 everywhere now (~Rs 4-7k whole studio), rung 4 on the bedroom + study windows first (sleep/work quality), skip rung 5 unless structural change happens anyway.

2. Textile defense

- **Triple-weave blackout curtains >=280 GSM:** SS Furnishings "Deam Out" 280 GSM satin triple-weave, Rs 685/panel (7 ft door size), machine-washable [37]. Warm neutral shade (taupe/greige family matches palette).
- Realistic acoustic expectation (honest): dense fabric absorbs ~30-35% of incident noise (high frequencies, echo) - takes the edge off street noise, does NOT block industrial rumble [39]. Combined with sealed windows it meaningfully improves sleep quietness.
- Wash cadence in this environment: quarterly (dust load).

3. Flooring decision

Winner: SPC rigid click-floor over existing tiles (living/bed/study/kitchen):

- Total installed **Rs 110-185/sqft standard / up to Rs 220 premium** [34][36]; zero demolition, zero cement-dust days (critical - renovation dust is exactly what we're fighting), laid in 1 day, walkable same day [34].
- 100% waterproof core (monsoon-safe), termite-proof, quieter underfoot than tile (attached IXPE underlay), warm-neutral feel, individual plank replacement possible [34][35].
- **Spec to demand:** 0.5 mm wear layer, 5-6 mm core, attached IXPE underlay; brands: Action Tesa, Mikasa (Greenlam), Pergo, Welspun, Square Foot [35]. Mid-tone oak-grey decor hides dust between sweeps.
- Runner-up: matte vitrified 600x600 (Rs 100-220 installed) - cheaper material but adds demolition (Rs 15-25/sqft), 3-5 days of mess, grout-mold maintenance in humidity [34]. Choose only if doing full rework anyway.
- Bathrooms: leave existing floor; anti-skid matters more there.

4. Sound + vibration kit

Item	Pick	Cost EST
Rug + pad (living)	low-pile flatweave + thick felt/rubber pad (pad does the damping)	Rs 2.5-5k
Acoustic wall panels	PET-felt panels behind bed headboard wall (NRC ~0.4-0.75 class)	Rs 250-600/panel x4-6
Appliance isolation	rubber anti-vibration pads under washing machine/fridge	Rs 300-600

Item	Pick	Cost EST
Wardrobe fixing	chemical anchors + neoprene washers (no metal-wall buzz path)	in joinery spec
Door	bottom sweep + seal	Rs 200-400

Expectation-setting: this kit + curtains + window seals ~ noticeable high-frequency/reverb reduction and buzz elimination; low factory hum persists. Don't oversell dB numbers - comfort gain is real but partial.

5. Air purification (soot control)

Sizing rule: for ~378 sqft x 9.5 ft ~ 3600 cu ft (~102 m3), 5 air changes/hr -> **CADR >= 350-400 m3/h** target.

- **Primary pick:** Xiaomi Smart Air Purifier 4 class (CADR ~400 m3/h, Rs 10-12k EST) or Philips AC1215 (CADR ~270, smaller room coverage) - verify current listing before purchase.
- **Budget backup (evidence-backed): DIY Corsi-Rosenthal box** - box fan + 4 MERV13 filters, Rs 4-6k build; peer-reviewed and university studies show performance rivalling commercial purifiers at a fraction of price; the agent track retrieved dedicated cost/performance studies [40]. Great as a second unit for the bedroom at night.
- Filter budget: genuine HEPA set annually Rs 1.5-3k; CR-box filters Rs 800-1200/yr.
- Behavioral: run purifiers on auto; keep windows closed during visible smog/dust hours; cross-ventilate briefly during clear morning windows.

[34] antoviasurfaces.com SPC vs vitrified 2026 - [35] studiomatrix.org SPC guide - [36] goalsfloors.com SPC cost breakdown - [37] ssfurnishings.com Deam Out 280GSM - [38] reddit.com/r/DIY acrylic window insert thread - [39] joydeco.com triple-weave spec (~35% noise absorption figure) - [40] descentanalysis.org Corsi-Rosenthal cost/performance study

06 - MASTER BUY LIST & Phased Budget (v1.0 - research-priced)

Status: all tracks merged. Every price = researched street EST (2026, INR); replace with local Shimoga/Bhadravati quotes before ordering. Sources: SOURCES.md.

Gates: nothing ordered before Phase 0 (signed site measure + physical sample approval).

A. Furniture & sleep - Rs 59-66k (recommended) / Rs 38-45k (essential)

Sofa Wakefit Napper Rs 20,299 - Mattress SleepyCat Latex Hybrid Rs 18-20k - Bed Solimo Petra sheesham Rs 14.5-17.5k - Chair Cellbell C104 Rs 2.9k - Desk Rs 3.5-6k - Bedding 2x Rs 800 - (details 01_furniture.md)

B. Joinery (kitchen shutters + 3-door wardrobe) - Rs 1.65-2.05L

Line	Spec	EST
Kitchen shutters-only (Sainik 710 core, S1241 Latte, Sensys hinges, ABS edges, labour)	~38 sqft shutter faces	Rs 55-65k
Wardrobe 70 sqft front (Sainik/HDHMR, Idria Oak 84689, Ebco soft-close, internals)	mid laminate spec Rs 1500-1800/sqft	Rs 1.05-1.4L
Anti-vibration wall fixing + mirror safety film	-	incl.

C. Paint system - Rs 18-24k

Berger Easy Clean Fresh walls Rs 346/L (~5L) + ceiling (~3L) ~ Rs 2.8k material - primer Rs 1.4k - putty 40kg Rs 1.2k - stain-block spot primer Rs 800 - labour ~1000 sqft @ Rs 12-18 ~ Rs 12-18k. Shades: NN9074 main / NN9242 dim zones / WW0005 ceiling / GG7140 accent (Birla Opus codes, tint at dealer).

D. Lighting & electrical - Rs 38-52k

Line	Pick	EST
8x downlight 3000K CRI90	Philips Meson Comfort 13W (alt: Havells Luna Rs 215 ea)	Rs 8-10k
Kitchen under-loft strip + driver + profile	3000K diffused	Rs 1.5-2.5k
Bath + lamps	IP44 fixture, study lamp, 2 bedside	Rs 3-5k
Wiring	Polycab FRLS 3x1.5 + 2x2.5 sqmm rolls	Rs 8-12k
Protection	Easy9 RCCB 25A/30mA Rs 1,786 + SPD EZ9L33620 T2 Rs 2-3.5k + 8-way DB + 6 MCBs	Rs 6.5-8k
Switches ~30 modules + plates	Legrand Myrius NextGen @Rs 55-111	Rs 4-6k
Electrician labour	-	Rs 6-10k

E. Utilities & envelope - Rs 95-135k (full) / Rs 75-100k (defer chimney+purifier)

Line	Pick	EST
Fan 1200mm BLDC	Atomberg Efficio Alpha Rs 2,699	Rs 2.7k
Geyser 15L	AO Smith Blue Diamond glass-lined	Rs 6.6-11.5k
Bath exhaust 150mm gravity flap	Havells/Crompton	Rs 1k
Chimney 60cm auto-clean (<i>deferable</i>)	Faber/Elica entry	Rs 8-12k
Purifier CADR>=350 + CR-box backup	Xiaomi 4 class + DIY	Rs 15-17k
Window seals rungs 1-3	EPDM+pile+MS polymer+sweeps	Rs 4-7k
Secondary glazing bedroom+study (<i>deferable</i>)	acrylic inserts	Rs 8-12k
Curtains 4x 280GSM triple-weave + rods	SS Furnishings Rs 685/panel	Rs 4.2k
SPC flooring 397 sqft installed	0.5mm wear, 5-6mm core, IXPE (Action Tesa/Mikasa class)	Rs 52-73k
Rug+pad, PET-felt panels, AV pads	-	Rs 5-9k

PHASE TOTALS (EST)

Tier	Scope	Total
Essential	joinery + paint + electrical + furniture + envelope basics (defer: secondary glazing, chimney, purifier, premium rungs)	~ Rs 2.9-3.6L
Recommended (target)	everything above + SPC flooring + purifier + full envelope	~ Rs 3.9-4.9L
Premium	+ Greenply Gold boards, IKEA sofa, Meson everywhere, DGU windows	~ Rs 5.5L+

Order of execution (inherits repo priority): Phase 0 verify -> wet/storage core (kitchen+wardrobe) -> paint -> electrical rough-in -> lighting -> flooring (SPC last, after dusty trades) -> envelope seals -> furniture -> soft goods.

Cross-checks for quotes (anti-scam)

- Carpenter quote must itemize: board brand/grade/thickness per IS mark, laminate code, edge thickness, hinge model, Rs /sqft basis (front area), GST separate [17].
- Plywood invoice must show IS:710 mark + CM/L number [15].
- Electrical: RCCB+SPD on incomer, earthing verified, licensed electrician sign-off.
- Paint: tinted-primer + 2 topcoat system written into quote; leftover 1L per shade.

[15] <https://home-run.co/blogs/homerun-knowledge-base/bangalore-plywood-price-comparison>

[17] <https://www.studiomatrix.org/guides/wardrobe-cost-guide-india>

01 - Paint & Color Science (research track R-Paint)

Status: COMPLETE - merged from subagent research + shade-card verification

Locked base: see 00_SPEC_LOCKED_BASE.md §2-4. This file adds the cross-brand comparison, chemistry, and human validation.

1. The decision that matters most: washability class

In an industrial-belt home the wall finish is a **maintenance surface**, not just a color surface. Soot + fine dust deposit on walls; cleaning frequency will be monthly wipe-downs. That makes **scrub resistance (cycles)** and **anti-fungal additive package** the two specs to buy for - sheen stays matte/low-sheen per locked concept.

Buy rule: >=5000 scrub cycles class, anti-fungal/algae additive, low-sheen. In Birla Opus ladder: **Calista Ever Wash** (premium pick) or **Style Color Smart** (economy pick). Luxury tier = One Pure Elegance Matt if budget allows.

2. Brand-series comparison (2025-26 street reality)

Brand	Series	Chemistry	Washability claim	Anti-fungal	Sheen	Street Rs /L*	Verdict
Birla Opus	Calista Ever Wash	acrylic emulsion	5000+ scrubs	yes	low-sheen/matt	~380-450	WINNER premium-value - matches card system notes
Birla Opus	Style Color Smart	emulsion	washable	yes	matt	~250-320	Winner economy - same shade library
Birla Opus	One Pure Elegance Matt	luxury acrylic	scuff+stain, 7-yr class	yes	matt	~600-750	only if splurging on living room
Asian Paints	Royale Aspira/Shyne	pure acrylic + additive pkg	high scrub	yes	soft-sheen	~550-800	best-known brand tax; comparable perf
Asian Paints	Apcolite Advanced	solvent-borne enamel-ish	very washable	n/a	satin	~450-600	overkill; stronger odor/VOC
Berger	Easy Clean / Silk	acrylic	stain-release tech	yes	low-sheen	~400-520	strong #2; check local tinting service
JSW Impex/Prestige	acrylic emulsion	mid scrub class	yes	matt	~300-400	value but thinner dealer/tinting net	
Dulux Velvet Touch	acrylic	high scrub	yes	velvet-matt	~500-650	good product, weaker presence in Shimoga belt	

*Prices = EST from retail aggregators + typical market rates; verify at local Shimoga/Bhadravati dealer before ordering. Tinting service quality matters as much as paint - Birla Opus dealers are new but aggressively priced; Asian Paints has the deepest dealer network in Karnataka small towns.

3. Primer & putty

Slot	Pick	Why
Putty	Acrylic-based white cement putty (Birla White Wallcare / Birla Opus One Pro), 2 coats	bonds to plaster, resists flaking vs pure acrylic-only putties on humid walls
Primer	Interior primer tinted toward NN9074 (Calista Pro White or One Pro)	tinted primer = fewer topcoats needed = saves money
Sealer (smoke)	if old smoke stains bleed, one coat of stain-blocking primer (Asian Paints Tractor Sealer class or oil-based primer spot-primer)	soot is oily; water-based alone can let it bleed through

4. Shade plan (locked + industrial-ladder refinement)

Room	Shade	Code	Logic
Living, kitchen walls, bedroom main	warm greige	NN9074 Puddle of Grey (LOCKED)	mid-tone hides soot streaks; harmonizes Latte+Idria
Bath corridor / wardrobe surround (dim zones)	lighter khaki-neutral	NN9242 A Khadi Kurta	factory-recommended "industrial areas" wall shade - brighter in dim light, still dust-tolerant
Ceiling all rooms	warm linen white	WW0005 White Linen (LOCKED); alt WW0020	card's industrial ceiling rec
Accent wall (living, max 1)	sage	GG7140 Tender Buds (optional)	calm, pairs with Idria green-grey grain
Study optional accent	dusty blue-green	BB5146 Old Novel	one controlled surface only

Sample pots of all four shades approved under morning sun, afternoon sun, 3000 K evening before bulk purchase (metamerism rule).

5. Human validation (real users)

- **Berger Easy Clean Fresh - WINNER (paint agent's full research pass):** Rs 346/L street, 20L tin Rs 6,924. Film = 3D webbed network of modified polyester fibres cross-linked with acrylic - resists stain penetration, hairline cracks, superior hiding; soft-smooth sheen (inside matte/low-sheen lock); anti-fungal built in; Berger dealer network covers Shimoga/Bhadravati.
- **Team-BHP owner (Bangalore project):** *"While we initially planned on taking the 'Silk Glamour' Paint, the SA advised us... its better to take the 'Easy Clean Fresh' due to its stain resistance and washability."* - dealers themselves route washability-seekers here [41].
- **MouthShut Aspira owner (cautionary):** *"if I need to repaint my wall then it would remove the colour... disgusting experience with Asian paints Royal Aspire."* - even premium films lift under aggressive scrubbing; wash gently regardless of brand [42].
- **NoBroker thread:** *"The paint resists stains well, but it needs gentle cleaning to keep its shine."* [43]
- Video head-to-head Easy Clean vs Royale [44].
- Skip list from the pass: Royale Shyne / JSW Halo Aquaglo / Dulux Pearl Glo (**gloss-banned**), Apex Ultima (EXTERIOR product), Tractor Emulsion (won't survive wash cycles), Dulux Super Cover Ultra (non-washable economy).

Agent comparison table (13 series)

Brand	Series	Chem	Sheen	Coverage sqft/L/coat	Rs /L incl tint
Berger	Easy Clean Fresh [OK]	acrylic + polyester cross-link	soft smooth	140-160	346
Berger	Silk Glamour Matt	pure acrylic, low VOC	matt-soft	150-170	512
Asian	Apcolite Advanced	100% acrylic	smooth matte	120-140	380-480
Asian	Royale Aspira	luxury + Teflon protector	matte/satin	160-180	700-900
Asian	Tractor Emulsion	vinyl-acrylic economy	dead matte	90-110	130-180
Birla Opus	Std emulsion	100% acrylic new-gen	matte	~150	250-320
Birla Opus	Pearl Glo	acrylic	soft sheen	150	350-450
JSW	Halo Aquaglo	acrylic	semi-gloss - skip	130-150	350-450
Dulux	Velvet Touch Pearl Glo	acrylic luxury	pearl - skip	140-160	475
Dulux	Super Cover Ultra	vinyl-acrylic economy	matte	100-120	193

VOC/scrub figures are water-based typical class - confirm printed TDS at dealer counter.

6. Primer & putty (agent verdict, upgrades §3)

- **Putty: white-cement based, NOT acrylic** - Birla White WallCare (~Rs 900-1,050/40 kg) or Birla Opus Wall Care Putty (Rs 588-668/10 kg). Cement putty is denser/less porous -> resists moisture uptake at 85% RH and survives repeated wet-wiping; acrylic putty stays soft and blisters in humid smoke-laden air (reserve for dent-filling only).
- **Primer: 1 coat water-based cement primer** (Opus Wall Primer Rs 180-240/L) + **solvent-based stain-blocking primer ONLY on soot patches** - never topcoat aged soot directly, it bleeds through.

- Application protocol: putty cures 48h+; walls dry 7 days post-monsoon before painting; sugar-soap wash -> dry -> spot-prime -> feather topcoat on soot zones.

6b. Shade scheme reconciliation

The agent proposed an Asian-Paints-code scheme (Country Beige 8600 / Cane Beige 8563 / Beige Rage 8770 / ceiling Basra Pearl-N 0974), Berger Color Bank can computer-match those codes. **However the LOCKED repo palette stays governing** (NN9074 walls / WW0005 ceiling / GG7140 accent - catalogue-verified in this repo with swatch-lock audit). Resolution: order NN9074/WW0005/GG7140 in the winning paint system (Berger Easy Clean Fresh via code-match against the Birla Opus shade card hex values, or buy Birla Opus Calista Ever Wash in native codes). If client prefers the agent's warmer scheme, it's a client decision - not a spec change.

7. Application notes for smoky/dusty environment

1. Cadence from the agent pass: microfiber dry-dust monthly; damp wipe quarterly (kitchen + switch zones); full mild-soap wash 2x/year - clean soot off before it ages >6 months (aged soot bonds permanently).
2. Two putty coats sanded fine -> tinted primer -> 2 topcoats.
3. Never: abrasive pads, bleach, solvents on emulsion (uneven sheen polishing).
4. Keep 1 L tinted leftover per shade for touch-ups.
5. Kitchen splash zone behind hob: ceramic tile dado instead of painted wall.
6. Envelope sealing (window beads + door sweep) halves the wash load more than any paint upgrade.

8. Provenance

- Shade codes/RGB + product ladder + industrial-area recommendations: repo-internal, catalogue-verified - processed_pdf_text/birla_opus_shade_card_extracted.txt (2,322 shades, compiled 03-Aug-2026; card system notes pp.1) and FINAL_DELIVERABLE/coordination/swatch_lock.json (hex/RGB match confirmed by the repo's own audit).
- Brand-series price table \$2: market EST [unverified] - replace with dealer quotes.
- Berger Easy Clean Fresh pick: subagent research pass (transcript_agents/work/research_raw/), price Rs 346/L [unverified - verify locally].

Sources rendered in SOURCES.md. Raw search evidence archived by research agent.

[41] <https://www.team-bhp.com/forum/shifting-gears/266851-berger-express-painting-review-bangalore.html>

[42] <https://www.mouthshut.com/product-reviews/asian-paints-royale-aspira-reviews-925866611>

[43] <https://www.nobroker.in/forum/asian-paints-royale-vs-berger-silk-which-is-better>

[44] <https://www.youtube.com/watch?v=cwya-Um0MgM>